THE INDUCTION STEP FOR A FINITELY PRESENTED FLAT RING MAP WITH GEOMETRICALLY INTEGRAL FIBRES

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file flat.tex; no statement or proof step was rewritten, reordered or removed.

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Allowed standard prerequisites (editorial).

- The algebraic induction step for a map of modules over such a ring map. Same chapter flat.tex, label lemma-induction-step, invoked at proof line 3181. Applied to the localised map alpha at r, it gives the equivalence of conditions (1), (3), (4), (5) and (6); the remaining implication (1) implies (2) is what the counted proof supplies.
- Openness of flatness, openness of the locus where a map of finitely presented modules is an isomorphism, and openness of a flat finitely presented ring map. Algebra, labels algebra-theorem-openness-flatness (proof line 3190), algebra-lemma-map-between-finitely-presented (proof line 3203) and algebra-proposition-fppf-open (proof line 3207). Makes the flat locus U_1, the isomorphism locus U_2 and the image Phi(U_1 intersect U_2) open in the appropriate spectra, which is what allows the choice of f in R and then of g in S.
- Testing universal injectivity on stalks. Algebra, label algebra-lemma-universally-injective-check-stalks, invoked at proof line 3218. Converts universal injectivity of alpha at every prime of the open set Phi(U_1 intersect U_2) into universal injectivity of the localised map alpha_f, which is the condition required by item (2).

Source reference. Original lemma 05GF: sources/stacks-chow/tags.txt, source lines 7059–7059. Navigation pointer for \ref{algebra-lemma-map-between-finitely-presented} used inside the counted span (line 3203); the argument that consumes it is retained in the span.

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Source reference. Original lemma 05CL: sources/stacks-chow/tags.txt, source lines 6924–6924. Navigation pointer for \ref{algebra-lemma-universally-injective-check-stalks} used inside the counted span (line 3218); the argument that consumes it is retained in the span.

Source reference. Original lemma 00I1: sources/stacks-chow/tags.txt, source lines 663–663. Navigation pointer for \ref{algebra-proposition-fppf-open} used inside the counted span (line 3207); the argument that consumes it is retained in the span.

Source reference. Original lemma 00RC: sources/stacks-chow/tags.txt, source lines 998–998. Navigation pointer for \ref{algebra-theorem-openness-flatness} used inside the counted span (line 3190); the argument that consumes it is retained in the span.

Source reference. Original lemma 05I3: sources/stacks-flat/pinned-original-source.tex, source lines 2356–2356. Navigation pointer for \ref{lemma-induction-step} used inside the counted span (line 3181); the argument that consumes it is retained in the span.

Lemma 0.1. *Let R be a ring. Let $R \to S$ be a finitely presented flat ring map with geometrically integral fibres. Let $\mathfrak{q} \subset S$ be a prime ideal lying over the prime $\mathfrak{r} \subset R$. Set $\mathfrak{p} = \mathfrak{r}S$. Let N be a finitely presented S-module. There exists $r \geq 0$ and an S-module map*

$$\alpha : S^{\oplus r} \longrightarrow N$$

such that $\alpha : \kappa(\mathfrak{p})^{\oplus r} \to N \otimes_S \kappa(\mathfrak{p})$ is an isomorphism. For any such α the following are equivalent:

1. *$N_\mathfrak{q}$ is R-flat,*
2. *there exists an $f \in R$, $f \notin \mathfrak{r}$ such that $\alpha_f : S_f^{\oplus r} \to N_f$ is R_f-universally injective and a $g \in S$, $g \notin \mathfrak{q}$ such that $\mathrm{Coker}(\alpha)_g$ is R-flat,*
3. *$\alpha_\mathfrak{r}$ is $R_\mathfrak{r}$-universally injective and $\mathrm{Coker}(\alpha)_\mathfrak{q}$ is R-flat*
4. *$\alpha_\mathfrak{r}$ is injective and $\mathrm{Coker}(\alpha)_\mathfrak{q}$ is R-flat,*
5. *$\alpha_\mathfrak{p}$ is an isomorphism and $\mathrm{Coker}(\alpha)_\mathfrak{q}$ is R-flat, and*
6. *$\alpha_\mathfrak{q}$ is injective and $\mathrm{Coker}(\alpha)_\mathfrak{q}$ is R-flat.*

Proof. To obtain α set $r = \dim_{\kappa(\mathfrak{p})} N \otimes_S \kappa(\mathfrak{p})$ and pick $x_1, \ldots, x_r \in N$ which form a basis of $N \otimes_S \kappa(\mathfrak{p})$. Define $\alpha(s_1, \ldots, s_r) = \sum s_i x_i$. This proves the existence.

Fix a choice of α. We may apply Lemma 05I3 to the map $\alpha_\mathfrak{r} : S_\mathfrak{r}^{\oplus r} \to N_\mathfrak{r}$. Hence we see that (1), (3), (4), (5), and (6) are all equivalent. Since it is also clear that (2) implies (3) we see that all we have to do is show that (1) implies (2).

Assume (1). By openness of flatness, see Algebra, Theorem 00RC, the set

$$U_1 = \{\mathfrak{q}' \subset S \mid N_{\mathfrak{q}'} \text{ is flat over } R\}$$

is open in $\mathrm{Spec}(S)$. It contains $\mathfrak{q}$ by assumption and hence $\mathfrak{p}$. Because $S^{\oplus r}$ and N are finitely presented S-modules the set

$$U_2 = \{\mathfrak{q}' \subset S \mid \alpha_{\mathfrak{q}'} \text{ is an isomorphism}\}$$

is open in $\mathrm{Spec}(S)$, see Algebra, Lemma 05GF. It contains $\mathfrak{p}$ by (5). As $R \to S$ is finitely presented and flat the map $\Phi : \mathrm{Spec}(S) \to \mathrm{Spec}(R)$ is open, see Algebra, Proposition 00I1. For any prime $\mathfrak{r}' \in \Phi(U_1 \cap U_2)$ we see that there exists a prime $\mathfrak{q}'$ lying over $\mathfrak{r}'$ such that $N_{\mathfrak{q}'}$ is flat and such that $\alpha_{\mathfrak{q}'}$ is an isomorphism, which implies that $\alpha \otimes \kappa(\mathfrak{p}')$ is an isomorphism where $\mathfrak{p}' = \mathfrak{r}'S$. Thus $\alpha_{\mathfrak{r}'}$ is $R_{\mathfrak{r}'}$-universally injective by the implication (1) $\Rightarrow$ (3). Hence if we pick $f \in R$, $f \notin \mathfrak{r}$ such that $D(f) \subset \Phi(U_1 \cap U_2)$ then we conclude that α_f is R_f-universally injective, see Algebra,

Lemma 05CL. The same reasoning also shows that for any $\mathfrak{q}' \in U_1 \cap \Phi^{-1}(\Phi(U_1 \cap U_2))$ the module $\mathrm{Coker}(\alpha)_{\mathfrak{q}'}$ is R-flat. Note that $\mathfrak{q} \in U_1 \cap \Phi^{-1}(\Phi(U_1 \cap U_2))$. Hence we can find a $g \in S$, $g \notin \mathfrak{q}$ such that $D(g) \subset U_1 \cap \Phi^{-1}(\Phi(U_1 \cap U_2))$ and we win. $\square$